

The Null Zoo: Size and Power of Backtest-Overfitting and Data-Snooping Tests on Synthetic Searches with Known Ground Truth

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Abstract

Corrections for backtest overfitting and data snooping—the deflated Sharpe ratio, the multiple-testing haircut, family-wise adjustments, White’s Reality Check and Hansen’s test for superior predictive ability (SPA)—are derived under assumptions that trading research violates. The Null Zoo is an open, versioned and pre-registered simulation benchmark that draws complete research searches (20 strategies, 504 daily returns) from nine return families with known ground truth and scores 15 validators by size and power, with 10,000 searches per cell. Hansen’s studentized SPA test, and Romano and Wolf’s StepM with it, rejected 4.3–12.6% of skill-less searches at a nominal 5%, while the unstudentized Reality Check rejected 4.5–6.4%; the pattern held on fresh seeds and in an independent second implementation. Pre-registered experiments trace the excess to the long-run variance estimate the bootstrap holds fixed, compounded under negative skew by the dependence between a mean and its estimated standard deviation: re-studentizing every resample restores calibration (5.0–6.2% outside the autocorrelated family), while the excess shrinks with the sample, grows with the number of strategies and reaches 17.8% at the block length the Python package arch uses by default. arch’s own SPA test is unaffected only because its `studentize` option does nothing. Where volatilities differ or most strategies are poor, a correctly studentized and recentered test is far more powerful than the Reality Check; a floored statistic with a strict inequality can reject when no strategy beats the benchmark. A lag-one autocorrelation of 0.2 raises the size of the single-series tests that ignore it to about 19.5%, and the deflated Sharpe ratio’s 0.95 rule misses 90.8% of searches containing a strategy with a true Sharpe ratio of 2. 7 of 9 pre-registered predictions held; all results reproduce from the released code.

1 Introduction

A researcher who tries many variants of a trading strategy and reports the best one reports a biased number. The best of N strategies with no skill has an in-sample Sharpe ratio that grows with N , roughly in proportion to $\sqrt{2\ln N}$ times the spread of the individual estimates [5], and the same selection effect pervades published factor research [18]. Two families of corrections are now standard. The first adjusts the best strategy’s own statistic: the deflated Sharpe ratio [4], built on the probabilistic Sharpe ratio [3]; the multiple-testing haircut of Harvey and Liu [17]; and family-wise adjustments such as those of Holm [20] and Šidák [34]. The second resamples the whole search: White’s Reality Check [37], applied to trading rules by Sullivan et al. [36], Hansen’s test for superior predictive ability [16] and the stepwise procedures of Romano and Wolf [31] and Hsu et al. [21].

Each correction rests on assumptions—normal or independent returns, independent trials, large samples—that real searches violate in predictable ways. Returns are fat-tailed, skewed, volatility-clustered and sometimes autocorrelated; strategy variants share most of their positions and so are correlated; and two years of daily data is a short sample for a variance estimate. Inference for a single Sharpe ratio under such violations is well studied [22, 23, 27]. How much each violation distorts each *selection* correction, applied to the best of a search as practitioners apply it, is rarely answered in a form a practitioner can look up.

This paper answers it with a simulation benchmark, the *Null Zoo*. It draws complete research searches—20 candidate strategies of 504 daily returns each—from nine return families whose ground truth is known by construction, applies 15 validators as their implementations are meant to be called, and records how often each one declares a skilled strategy when there is none (its *size*) and when one strategy has a true annualized Sharpe ratio of 1, 2 or 3 (its *power*). The main run has 10,000 searches per cell. Every governed run was pre-registered before it was run, following recent arguments that comparative simulation studies need the same protection against selective reporting as empirical ones [28, 35]; a surprising result was confirmed on fresh seeds under a new pre-registration before it was used, and its explanation was tested by an independent implementation in a second language.

The findings:

1. **Hansen’s studentized SPA test, and StepM with it, is oversized at this horizon; White’s Reality Check is not.** At a nominal 5%, the consistent SPA test rejected 4.3–12.6% of skill-less searches across families and runs, the Reality Check 4.5–6.4%.
2. **The cause is a variance estimate the bootstrap holds fixed.** The SPA statistic divides each mean by a long-run standard deviation estimated once and reused in every resample, so the bootstrap ignores the estimate’s noise; under negative skew the mean and its estimated standard deviation are also dependent. Re-studentizing inside the bootstrap removes the excess (5.0–6.2% outside the autocorrelated family), as Hall and Wilson’s guideline would predict [15]. The excess shrinks with the sample (9.5% with one year of i.i.d. data, 6.2% with ten), grows with the number of strategies and with the block length, and reaches 17.8% under negative skew at the block length the Python package arch uses by default.
3. **arch’s SPA test stays close to its level because it does not studentize.** In arch 8.0.0 [33] the SPA test’s `studentize` option has no effect on the result; a pending change [1] would, on our implementation of its formula, make it oversized at two years of daily data.
4. **Studentizing and recentring still pay, when done right.** When volatilities differ, the unstudentized Reality Check almost never finds a skilled low-volatility strategy (power 5.0% against 53.1% for the bootstrap-*t* SPA test); when most strategies are clearly poor, consistent recentring raises power from 51.4% to 87.3%. A floored SPA statistic with a strict inequality can reject when no strategy beats the benchmark; counting ties fixes it.
5. **Autocorrelation breaks the single-series tests that ignore it.** A lag-one autocorrelation of 0.2 raises their false-positive rate to about 19.5%; Lo’s factor removes the excess in that case, and kernel and block-bootstrap remedies reduce it.
6. **The deflated Sharpe ratio with its authors’ 0.95 rule tests a harder hypothesis than “no skill”** and fails to accept 90.8% of i.i.d. searches containing a strategy with a true Sharpe ratio of 2 over two years; most of that is the threshold, not the statistic.

7. **Selection, not correction, sets the ceiling.** An oracle that knows which strategy is skilled rejects in 87.7% of such searches; the best calibrated best-of-20 tests reach about 52.9%.

The mechanism in finding 2—a noisy long-run variance inflating a studentized test—is familiar from tests of equal forecast accuracy [11, 19]. What the benchmark adds is its size where strategy research operates, its interaction with the maximum over strategies and with skewness, a remedy that keeps the benefits of studentizing, and its consequences for widely used code.

The benchmark scores a statistic introduced by its own author, the *luck-equivalent-trials* count, and the v1 runs were also used to choose a default test for the author’s open-source software. Bartoš et al. [7] argue that simulation studies designed by a method’s developers carry a conflict of interest. We report that conflict, pre-register the runs, publish every generator, validator, seed and result file, and note that the benchmark’s clearest failures—autocorrelation, and the floored strict-inequality p -value—apply to the author’s own implementations as much as to anyone’s.

Related work. Arian et al. [2] compare out-of-sample testing schemes—walk-forward, k -fold, purged and combinatorial purged cross-validation—in a synthetic market with known structure, scoring them by the probability of backtest overfitting [6] and the deflated Sharpe ratio. Their object is the scheme that produces a backtest; ours is the correction applied to its result. Santoni et al. [32] combine the deflated Sharpe ratio, the probability of backtest overfitting, the SPA test and the minimum track-record length into one robustness grade and report its discrimination on synthetic markets; they do not report the size of the individual components at a nominal level, which our results suggest matters for the SPA component in particular. Hsu et al. [21] and Hansen [16] evaluate the SPA test and its stepwise extension by simulation in their own designs, and Frederiksen et al. [13] study predictive-ability tests when losses have infinite variance. The size and power of multiple-testing procedures under dependence are well studied in statistics; our contribution is a small, open, versioned and pre-registered benchmark that uses the return shapes and search structures of trading research and calls each correction the way it is called in practice.

2 Design

We plan and report the study with the ADEMP structure of Morris et al. [25]: aims, data-generating mechanisms, estimands, methods and performance measures.

2.1 Aims

For each validator and each return family the study estimates (i) the probability that the validator calls the best of N skill-less strategies skilled (its *size*), and (ii) the probability that it rejects when one of the N strategies has real skill (its *power*). A validator is *calibrated* in a family when its size is within Monte Carlo error of its nominal level.

2.2 Data-generating mechanisms

One replication draws a *search*: $N = 20$ return series of $T = 504$ periods, read as two years of daily returns ($q = 252$ periods per year). Every series has population mean zero and population variance one, so every strategy’s true Sharpe ratio is zero. In the skill arms, series 0 receives a constant drift of $s/\sqrt{252}$ per period, which gives it a true annualized Sharpe ratio of $s \in \{1, 2, 3\}$; the other series are unchanged. The nine families are:

IID normal. Independent standard normal returns; the case every correction is derived for.

Student t_4 . Student t with 4 degrees of freedom, scaled to unit variance: symmetric fat tails. Its fourth moment is infinite, so sample kurtosis does not converge.

Skew ± 1.32 . A centred, unit-variance lognormal with log-standard deviation 0.4, whose skewness is +1.32, and its mirror image (-1.32).

GARCH(1,1). $h_t = 0.05 + 0.10 \varepsilon_{t-1}^2 + 0.85 h_{t-1}$, $\varepsilon_t = \sqrt{h_t} z_t$ [9], with unconditional variance one; each series starts at $h = 1$, $\varepsilon = 0$, without burn-in.

AR(1). $x_t = 0.2 x_{t-1} + \sqrt{1 - 0.2^2} z_t$, started from its stationary distribution.

Two volatility regimes. Normal returns whose standard deviation is 0.50 or 1.32 (0.6 and 1.6 rescaled to unit variance), switching with probability 0.02 each period.

Correlated trials. $x_{k,t} = \sqrt{1/2} f_t + \sqrt{1/2} z_{k,t}$ with a factor f_t common to all strategies, so every pair of strategies has correlation 0.5.

Block clusters. Four independent clusters of five strategies; strategies in a cluster share a factor with pairwise correlation 0.8. Twenty strategies that are really about four ideas, the shape a parameter grid around a few rules produces.

Instrument tests run before any validator is scored: every family's sample mean, variance, skewness, lag-one autocorrelation, autocorrelation of squared returns and cross-strategy correlation are checked against the property it claims, and the drift is checked to give series 0, and only series 0, the intended Sharpe ratio.

2.3 Estimands

In each replication the validator sees all 20 series and returns a p -value for the hypothesis that the best strategy has no skill; the single-series validators select the strategy with the highest sample Sharpe ratio, the joint tests consider all strategies at once. The estimands are the rejection rates $\Pr(p \leq 0.05)$ in the null arm (size) and the skill arms (power). Because raw power rewards a liberal test, we also report size-adjusted power: the power at the cutoff at which the same validator rejects at most 5% of the matching null cell. A user cannot apply that cutoff without knowing the family, but it shows how well each statistic ranks skilled searches above skill-less ones.

2.4 Methods

Let $\widehat{SR}$ be the best strategy's per-period Sharpe ratio (sample standard deviation with divisor $T-1$), $\widehat{SR}_a = \widehat{SR} \sqrt{q}$ its annualized value, and $\hat{\gamma}_3, \hat{\gamma}_4$ its sample skewness and (non-excess) kurtosis. The single-series validators are:

LET. The luck-equivalent-trials test. Under the null and normal returns, $t = \widehat{SR} \sqrt{T}$ is Student t with $T-1$ degrees of freedom, so one skill-less strategy reaches the observed value with probability $p_1 = \Pr(t_{T-1} \geq t)$, and the best of N independent ones with probability $1 - (1 - p_1)^N$ [34], the reported p -value. As a test it is the Šidák-adjusted Student t test; the statistic reports the same evidence in units of trials, $N_q = \ln(1 - q) / \ln(1 - p_1)$.

LET-Lo. LET after deflating $\widehat{SR}_a$ by the autocorrelation-adjusted annualisation factor of Lo [23], $[1 + \frac{2\hat{\rho}}{1-\hat{\rho}}(1 - \frac{1-\hat{\rho}^q}{q(1-\hat{\rho})})]^{-1/2}$, at the best series' lag-one autocorrelation $\hat{\rho}$ (clamped to $[-0.95, 0.95]$), as the haircut code of Harvey and Liu [17] does. The factor assumes $\rho_k = \rho^k$, which the AR(1) family satisfies exactly.

LET-NN. LET with the normal approximation and the non-normal standard error of Mertens [24] evaluated at the estimate: $z = \widehat{SR}\sqrt{T-1} / \sqrt{1 - \hat{\gamma}_3\widehat{SR} + \frac{\hat{\gamma}_4-1}{4}\widehat{SR}^2}$.

HC, HC-Lo, Holm. The haircut Sharpe ratio of Harvey and Liu [17]: the best strategy’s two-sided Student t p -value, adjusted by Bonferroni for 20 tests, without and with the Lo factor, and the Holm [20] variant. Harvey and Liu define the haircut with two-sided p -values, which for a positive Sharpe ratio is a one-sided test at 2.5%; to compare tests at a common level we also read it one-sided (two-sided p halved), and the tables report that reading unless stated.

DSR. The deflated Sharpe ratio [4]: the probabilistic Sharpe ratio [3] against a benchmark equal to the expected maximum Sharpe ratio of N skill-less trials, computed with the best strategy’s skewness and kurtosis and the cross-trial standard deviation of the 20 annualized Sharpe ratios. Its authors accept a strategy when $DSR \geq 0.95$; we score $1 - DSR$ as its p -value, so that rejection at 5% is exactly that rule.

Boot. An i.i.d. bootstrap [12] of the best strategy’s demeaned returns with $B = 1,999$ resamples; the single-trial p -value $(h + 1)/(B + 1)$, where h counts resampled Sharpe ratios at or above the observed one, is Šidák-adjusted for 20 trials.

HAC- t . The best strategy’s mean divided by a Newey–West (Bartlett kernel) standard error [26] with lag $\lfloor 4(T/100)^{2/9} \rfloor$, read against Student t with $T - 1$ degrees of freedom and Šidák-adjusted.

Boot- t . A stationary-bootstrap t test of the best strategy’s mean [29] with mean block length $\text{round}(T^{1/3})$ and 4,999 resamples, each resample studentized by its own standard deviation; Šidák-adjusted.

The joint resampling tests use all 20 series and one stationary bootstrap of the whole search (the same resampled periods for every strategy, so the dependence between strategies is kept), with mean block length $b = \text{round}(T^{1/3}) = 8$ and 1,000 resamples. With $\bar{d}_k$ the mean of strategy k and $\bar{d}_k^*$ its mean in a resample:

RC. White’s Reality Check [37]: $\max_k \bar{d}_k$ against the resampled distribution of $\max_k (\bar{d}_k^* - \bar{d}_k)$. It is not studentized.

SPA_c, SPA_u. Hansen’s test for superior predictive ability [16]: $T^{\text{SPA}} = \max(0, \max_k \sqrt{T} \bar{d}_k / \hat{\omega}_k)$, where $\hat{\omega}_k^2 = \hat{\gamma}_{0,k} + 2 \sum_{i=1}^{T-1} \kappa_i \hat{\gamma}_{i,k}$, $\kappa_i = (1 - \frac{i}{T})(1 - \frac{1}{b})^i + \frac{i}{T}(1 - \frac{1}{b})^{T-i}$, is the Politis–Romano estimate of the long-run variance at the bootstrap’s block length; the resampled statistic is $\max(0, \max_k \sqrt{T}(\bar{d}_k^* - g_k) / \hat{\omega}_k)$ with the same $\hat{\omega}_k$. The consistent p -value recentres at $g_k = \bar{d}_k$ when $\bar{d}_k \geq -\hat{\omega}_k \sqrt{2 \ln T / T}$ and at zero otherwise; the upper p -value recentres every strategy at its mean.

StepM. The stepwise procedure of Romano and Wolf [31] on the same studentized statistics and consistent recentring; it returns a set of strategies rather than a p -value, so we score whether it declares at least one strategy superior at 5%.

Finally, an *oracle* knows which strategy is skilled and applies a one-sided Newey–West t test to it alone. No valid best-of- N test can be more powerful, so the oracle’s power is a ceiling for every other column.

LET, HC, HC-Lo and DSR call the implementations behind the author’s public calculator and API, and RC, SPA and StepM the data-snooping implementation those use; the rest are written in the benchmark. No validator is tuned to the benchmark.

2.5 Performance measures, Monte Carlo error and runs

Each cell (family, skill level) of the main run, Null Zoo v1, has 10,000 searches. The Monte Carlo standard error of a rejection rate $\hat{p}$ from n searches is $\sqrt{\hat{p}(1-\hat{p})/n}$: 0.22 percentage points at 5% for $n = 10,000$. We flag a size as miscalibrated when it lies outside the 99% band around 5% (4.44–5.56% for 10,000 searches), so that a calibrated validator is flagged in about one size cell in a hundred.

The study has three governed runs, each pre-registered before it was run:

- **v1** (base seed 20261004). Its pre-registration fixed the families, the skill levels, the number of searches, a size bar (at most 6% at a nominal 5% in every family), a power bar (at least 45% at a Sharpe ratio of 2 under i.i.d. returns), the oracle ceiling and an ordered list of candidate tests, to choose a default test for the author’s open-source validation software. The listed joint-test candidates were `spa_consistent`, `spa_upper`. The Reality Check was measured but not listed.
- **v1b** (base seed 20261005). v1 found the Reality Check calibrated where the listed SPA candidates were not. Adopting it on that evidence would have been a choice made after seeing the results, so a separate pre-registration added it to the list and repeated the run on fresh seeds; v1b’s result decided the default whichever way it fell.
- **v2** (base seed 20261009; commit 7658b3b46ddb of the repository). An independent implementation in a second language, written from the published definitions rather than translated from v1’s code, tests why the studentized SPA test is oversized and where studentizing pays off. Its pre-registration fixed four experiments and 9 predictions; a pilot run on another seed, which shaped the design, is disclosed in the pre-registration and shares no search with the governed run. Section 4 reports every prediction, including any that failed.

v1 and v1b use a xoshiro128** generator [8] with one seed per chunk of searches, merged in a fixed order so the output is identical for any number of worker threads; v2 uses numpy’s PCG64 generator in the same way. An earlier published version (v0, 2,000 searches per cell with seven validators and eight families) and its 10,000-search robustness run are reported in the first version of this paper; v1 supersedes their size and power estimates, and v0 still reproduces byte for byte from its code.

3 Results: single-series corrections

Tables 1 and 2 report size in v1, Table 4 raw and size-adjusted power at a Sharpe ratio of 2, and the appendix the remaining validators, the other skill levels and the confirmation run v1b. Rates in the text are from v1 unless stated.

3.1 The baseline is calibrated, and the search itself costs power

Under independent normal returns the luck-equivalent-trials test is exact by construction: the t -statistic is Student t and the trials are independent. Its size is 4.9% (5.3% in v1b). The non-normal-error version (4.9%), the one-sided haircut (4.8%), the bootstrap (4.8%) and the Reality Check (5.1%) are calibrated too. The Newey–West tests are slightly liberal even here (HAC- t 5.6%, the single-strategy oracle 5.6%): a kernel estimate of the variance adds noise to the denominator of t , a theme Section 4 returns to.

Table 1: Size of the single-series corrections: percentage of skill-less searches (20 strategies, 504 daily returns) whose best strategy is called skilled at the nominal 5% level. Null Zoo v1, 10,000 searches per cell.

Family	LET	LET-Lo	LET-NN	HC	DSR	Boot	HAC- <i>t</i>	Boot- <i>t</i>
IID normal	4.9	5.1	4.9	4.8	<i>0.1</i>	4.8	5.6	6.1
Student t_4	4.8	5.2	6.7	4.6	<i>0.1</i>	6.3	5.7	7.9
Skew -1.32	8.3	8.7	<i>4.4</i>	8.1	<i>0.1</i>	5.5	9.0	6.7
Skew $+1.32$	<i>3.0</i>	<i>3.4</i>	6.1	<i>2.9</i>	<i>0.1</i>	4.9	<i>3.7</i>	6.6
GARCH(1,1)	4.9	5.0	5.1	4.8	<i>0.1</i>	4.9	5.3	6.0
AR(1), $\phi = 0.2$	19.5	5.3	19.4	19.1	<i>0.3</i>	18.4	7.4	7.6
Two vol. regimes	5.0	5.2	5.5	4.9	<i>0.1</i>	5.4	5.1	6.0
Correlated, $\rho = 0.5$	<i>3.5</i>	<i>3.8</i>	<i>3.6</i>	<i>3.5</i>	<i>1.7</i>	<i>3.3</i>	<i>4.0</i>	<i>4.3</i>
Clusters, $\rho = 0.8$	<i>3.3</i>	<i>3.5</i>	<i>3.3</i>	<i>3.2</i>	<i>0.0</i>	<i>3.2</i>	<i>3.7</i>	<i>3.9</i>

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it. HC is the Bonferroni haircut read one-sided (two-sided p halved); Boot uses 1,999 resamples and Boot- t 4,999.

Table 2: Size of the joint resampling tests in Null Zoo v1 and in its confirmation run v1b (fresh seeds), 10,000 searches per cell each, 1,000 stationary-bootstrap resamples.

Family	RC v1	RC v1b	SPA _c v1	SPA _c v1b	SPA _u v1	StepM v1
IID normal	5.1	5.4	7.5	7.8	7.4	7.5
Student t_4	4.9	4.9	7.3	7.6	7.2	7.3
Skew -1.32	5.4	5.3	12.5	12.6	12.2	12.5
Skew $+1.32$	4.9	5.2	<i>4.3</i>	<i>4.3</i>	<i>4.2</i>	<i>4.3</i>
GARCH(1,1)	4.8	4.5	6.5	6.3	6.4	6.5
AR(1), $\phi = 0.2$	6.3	6.4	8.9	9.1	8.8	8.9
Two vol. regimes	4.8	5.1	5.7	6.2	5.5	5.7
Correlated, $\rho = 0.5$	5.6	5.5	7.1	6.9	7.1	7.1
Clusters, $\rho = 0.8$	5.5	5.7	6.9	7.0	6.8	6.9

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it. StepM is scored as a decision (at least one strategy declared superior at 5%).

Selection, not the correction, costs most of the power. With a true Sharpe ratio of 2 over two years, the oracle that tests the skilled strategy alone rejects in 87.7% of searches; the best calibrated best-of-20 tests reach 52.9% (LET) and 53.5% (RC). The skilled strategy is the in-sample best in only 80.3% of these searches, so even a perfect correction would crown the wrong strategy in about one search in five; under AR(1) returns it is the best in 67.0%.

3.2 Autocorrelation breaks the tests that ignore it

With a lag-one autocorrelation of 0.2, the tests that treat returns as independent rejected about four times as often as their level: 19.5% for the luck-equivalent-trials test, 19.1% for the one-sided haircut, 19.4% with the non-normal standard error and 18.4% for the i.i.d. bootstrap. Positive autocorrelation makes the sample Sharpe ratio more variable than the i.i.d. formula assumes; the non-normal correction adjusts for the shape of the marginal distribution, which is normal here; and the i.i.d. bootstrap destroys the very dependence that inflates the Sharpe ratio. The deflated Sharpe ratio, which also ignores autocorrelation, stays far below its level (0.3%) only because it rejects almost nothing anywhere.

Table 3: Why the studentized SPA test is oversized: size of variants of the consistent SPA test on the same searches (Null Zoo v2, experiment A; independent Python implementation, 10,000 searches per cell, block length 8 unless marked 22).

Family	SPA _c	SPA _c -SD	SPA _c -BT	RC	SPA _c -U	SPA _c (22)	SPA _c -U (22)
IID normal	7.5	5.5	5.2	5.1	5.3	12.9	5.8
Student t_4	6.9	5.1	5.1	4.5	4.6	11.9	5.6
Skew -1.32	12.3	10.2	5.4	5.2	5.5	17.8	5.7
Skew $+1.32$	<i>4.2</i>	<i>2.8</i>	5.4	5.1	5.1	8.6	6.0
GARCH(1,1)	6.6	5.1	5.1	4.9	5.1	11.3	5.4
AR(1), $\phi = 0.2$	9.4	6.9	6.6	6.1	6.2	13.5	6.3
Two vol. regimes	5.8	4.8	5.0	4.5	4.8	9.3	5.0
Correlated, $\rho = 0.5$	6.7	5.6	5.5	5.4	5.4	10.2	6.3
Clusters, $\rho = 0.8$	7.6	6.4	6.2	6.1	6.2	11.4	7.3

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it. SPA_c: Hansen’s statistic studentized by a fixed Politis–Romano long-run variance. SPA_c-SD: studentized by the sample standard deviation instead. SPA_c-BT: every resample re-studentized by its own standard deviation (bootstrap- t). SPA_c-U: not studentized, the computation arch 8.0.0 performs. (22): block length 22, arch’s default for 504 periods.

Deflating the t -statistic by Lo’s annualisation factor removes the excess here: the luck-equivalent-trials test falls to 5.3% (5.5% in v1b). But the AR(1) family is the factor’s best case—it satisfies the assumed geometric decay exactly—and in the eight families without autocorrelation the factor shifts the size by 0.10–0.47 points, because the estimated factor adds noise to the statistic. The remedies that make no parametric assumption about the dependence reduce the excess without removing it at this sample size: a Newey–West standard error (HAC- t , 7.4%) and a stationary-bootstrap t test (Boot- t , 7.6%), and the joint resampling tests, whose blocks carry the dependence, are close but above the band (RC 6.3%, 6.4% in v1b).

3.3 Skew and fat tails

The Student- t tests are liberal under negative skew (8.3% for the luck-equivalent-trials test, 8.1% for the one-sided haircut, 9.0% for HAC- t) and conservative under positive skew (3.0%). With $T = 504$, the t -statistic of a skewed series is itself skewed: under negative skew a high sample mean tends to come with a low sample variance, so the upper tail of t is heavier than Student t , and the reverse under positive skew.

The non-normal standard error of Mertens [24] moves the error rather than removing it: it is conservative under negative skew (4.4%) and liberal under positive skew (6.1%) and fat tails (6.7%). It was not designed to correct the null distribution: at a Sharpe ratio of zero its variance term equals one for every i.i.d. distribution, and skewness and kurtosis enter only because the term is evaluated at the estimate. Selection then makes that estimate misleading. Under t_4 returns, a symmetric distribution, the selected best series has a mean sample skewness of 0.58: a series tends to win because it had a large positive outlier. The single-strategy bootstrap tests inherit the same selection effect (Boot 6.3%, Boot- t 7.9% under t_4): resampling a winner that owes its rank to an outlier reproduces the outlier’s influence. No single-series validator is calibrated across negative skew, positive skew and fat tails; the Reality Check, which resamples all strategies jointly, is (5.4%, 4.9% and 4.9%).

Under GARCH volatility clustering and regime switching the Student- t tests stay near 5% (LET 4.9% and 5.0%): returns remain a martingale difference sequence with finite variance, and two years

Table 4: Power when one strategy has a true annualized Sharpe ratio of 2 (raw / size-adjusted), Null Zoo v1. The oracle test knows which strategy is skilled and tests it alone; no valid best-of-20 test can beat it.

Family	Oracle	LET	LET-Lo	DSR	Boot	HAC- t	RC	SPA _c
IID normal	87.7 / 86.7	52.9 / 53.2	52.9 / 52.7	9.2 / 40.4	51.2 / 51.2	54.0 / 52.4	53.5 / 53.2	56.9 / 50.2
Student t_4	88.7 / 88.1	54.0 / 54.6	54.0 / 53.5	10.3 / 41.9	53.3 / 45.8	55.5 / 53.5	52.0 / 52.0	58.2 / 51.7
Skew -1.32	87.5 / 84.9	55.3 / 47.1	55.4 / 46.4	7.7 / 39.2	46.4 / 42.8	56.1 / 46.0	56.1 / 54.8	61.7 / 44.6
Skew $+1.32$	89.3 / 90.0	52.6 / 60.0	53.1 / 58.6	11.8 / 43.0	58.2 / 58.2	53.6 / 57.6	51.4 / 51.7	54.3 / 56.4
GARCH(1,1)	87.7 / 87.6	54.7 / 54.9	54.8 / 54.8	11.3 / 42.9	53.2 / 53.2	55.6 / 54.6	51.5 / 51.8	58.4 / 53.8
AR(1), $\phi = 0.2$	77.3 / 75.1	60.3 / 35.4	35.4 / 34.2	8.0 / 29.8	58.4 / n/a	40.3 / 34.3	38.5 / 34.8	42.5 / 32.4
Two vol. regimes	87.9 / 87.4	55.8 / 55.6	55.5 / 54.9	11.7 / 43.5	54.5 / 51.3	56.7 / 56.1	50.4 / 50.8	58.6 / 56.5
Correlated, $\rho = 0.5$	88.1 / 87.5	51.8 / 56.8	51.6 / 55.8	24.7 / 40.8	50.4 / 54.9	52.3 / 56.1	59.2 / 56.9	61.0 / 54.2
Clusters, $\rho = 0.8$	88.4 / 88.1	51.9 / 58.9	51.9 / 58.1	15.5 / 51.3	50.2 / 57.0	52.9 / 58.1	59.9 / 58.7	62.1 / 55.8

Size-adjusted power is power at the cutoff at which the same test rejects at most 5% of the matching null cell; a user who does not know the return family cannot apply it, but it shows how well each statistic ranks skilled searches above skill-less ones. n/a: the test’s smallest attainable p -value already occurs in more than 5% of the null cell, so no cutoff attains the level. Monte Carlo standard errors are at most 0.5 points.

Table 5: Size against the number of periods T and strategies k (Null Zoo v2, experiment B, 5,000 null searches per cell; block length $b = \text{round}(T^{1/3})$).

Design	IID normal				Skew -1.32			
	SPA _c	SPA _c -SD	SPA _c -BT	RC	SPA _c	SPA _c -SD	SPA _c -BT	RC
$T = 252, k = 20$ ($b = 6$)	9.5	6.0	5.5	5.2	16.2	12.7	5.8	5.0
$T = 504, k = 20$ ($b = 8$)	7.1	5.6	5.2	5.0	12.1	10.0	5.0	5.1
$T = 1,260, k = 20$ ($b = 11$)	7.1	6.2	6.0	5.8	9.1	7.9	5.1	5.1
$T = 2,520, k = 20$ ($b = 14$)	6.2	5.6	5.5	5.4	7.7	6.9	4.9	4.7
$T = 504, k = 5$ ($b = 8$)	6.9	6.3	6.2	6.0	8.7	8.0	5.4	5.7
$T = 504, k = 100$ ($b = 8$)	9.5	5.3	4.9	4.8	18.4	12.3	4.3	3.9

Bold: above the 99% Monte Carlo band around 5% (4.21–5.79 for 5,000 searches); italic: below it.

of daily data are enough for the t -test to be nearly exact.

3.4 Correlated strategies make the Bonferroni-type tests conservative

When every pair of strategies has correlation 0.5, or strategies come in clusters of five with correlation 0.8, the search contains fewer than 20 effectively independent trials, and every test that assumes 20 is conservative: 3.5% and 3.3% for the luck-equivalent-trials test, 3.3% and 3.2% for the bootstrap. The unused level costs power (LET 51.8% raw against 56.8% size-adjusted in the correlated family). The Reality Check, which resamples the strategies together and so sees their correlation, keeps its level (5.6% and 5.5%).

3.5 The deflated Sharpe ratio at 0.95 tests a harder hypothesis

Bailey and López de Prado [4] accept a strategy when $\text{DSR} \geq 0.95$. Scored this way, the deflated Sharpe ratio rejects at most 1.7% of skill-less searches (in the correlated family) and fails to accept 90.8% of i.i.d. normal searches that contain a strategy with a true Sharpe ratio of 2 over two years (75.3–92.3% across families; power 9.2% against 52.9% for the Šidák test).

Formally, the rule tests $H_0: SR \leq SR_0$, where SR_0 is the expected maximum of N skill-less trials, a harder hypothesis than “the best strategy has no skill”. Bailey and López de Prado [4]

Table 6: Where studentizing and recentring matter (Null Zoo v2, experiment C; i.i.d. normal returns, 504 periods, 20 strategies, 10,000 searches per cell). Size / power / size-adjusted power, in percent.

Test	Unequal volatilities	Poor alternatives
RC	5.5 / 5.0 / 4.4	0.2 / 51.4 / 88.9
SPA _c -U	5.7 / 5.2 / 4.3	4.8 / 87.3 / 87.7
SPA _c	7.6 / 56.7 / 48.9	7.4 / 87.4 / 82.7
SPA _u	7.5 / 56.4 / 49.0	0.3 / 53.3 / 88.3
SPA _c -SD	5.6 / 53.8 / 52.2	7.4 / 87.3 / 82.4
SPA _c -BT	5.2 / 53.1 / 52.4	7.3 / 87.1 / 82.2
LET	4.8 / 52.4 / 53.0	0.2 / 50.7 / 88.8

Unequal volatilities: volatilities from 0.5 to 2, all Sharpe ratios 0 under the null; under the alternative the least volatile strategy has a Sharpe ratio of 2. Poor alternatives: unit volatility, 19 strategies with a Sharpe ratio of -3 ; strategy 0 has 0 (null) or 2 (alternative). LET is the Šidák-adjusted t test on the best Sharpe ratio.

read the DSR as the probability that the true Sharpe ratio exceeds zero; read that way, a value of at least 0.95 should occur in about 5% of skill-less searches, so at its 0.95 point it is strongly conservative. The difference in hurdles is large:

- Under independence and normality, the one-sided 5% critical value for the best of $N = 20$ trials is about 2.80 standard errors.
- The DSR’s benchmark is about 1.90 cross-trial standard deviations (the approximation of Bailey et al. [5]; the exact expected maximum of 20 standard normals is 1.87), and the rule asks for a further 1.645 standard errors on top of that.
- The cross-trial standard deviation is computed from all 20 strategies, including the best. In the v0-design robustness run it averaged 0.99 times the i.i.d. standard error under the null, a hurdle of about 3.53 standard errors; under the alternative the skilled strategy widens the dispersion to 1.17 times the standard error and the hurdle to about 3.87: a skilled strategy raises its own benchmark.
- When strategies are strongly correlated the dispersion shrinks; below 0.61 times the standard error the DSR’s hurdle falls under the Šidák value. In the correlated family it was 0.70, a hurdle of about 2.97, which is why the DSR rejects most there.

The ranking the DSR induces is better than its raw power suggests: size-adjusted, its power under i.i.d. returns is 40.4% against 53.2% for the Šidák test, and across families it trails by 5.6–17.0 points. The 0.95 threshold costs most of the power; the statistic, whose benchmark moves with the dispersion of the whole search, costs the rest.

3.6 Resolution matters for the bootstrap

A bootstrap p -value with B resamples is a Monte Carlo estimate, and a test built on it loses power as B falls [10]. With a Šidák adjustment the loss is sharp: with 400 resamples the smallest attainable adjusted p -value is 0.0487, just below the level, so the test rejects only when no resample reaches the observed Sharpe ratio. Scoring the same 10,000 searches per cell with 400 and with 1,999 resamples (Table 12), the bootstrap’s power under i.i.d. normal returns rises from 46.7% to 50.9% (the Šidák t test: 52.6%), while its size moves from 4.28% to 4.50%. Comparisons of resampling tests with analytical ones should state the number of resamples.

3.7 The haircut read one-sided is the Šidák test

For a positive Sharpe ratio, the two-sided $p \leq 0.05$ of Harvey and Liu [17] is a one-sided test at 2.5%, which is why the haircut with its authors' convention sits near half the level (2.2% under i.i.d. normal returns). Read one-sided, the haircut and the luck-equivalent-trials test use the same Student t tail p_1 of the same statistic; Bonferroni rejects when $Np_1 \leq 0.05$ and Šidák when $1 - (1 - p_1)^N \leq 0.05$, and the two disagree only when p_1 lies between 0.002500 and 0.002561. Their sizes differ by at most 0.36 points in any family. This is a known equivalence, not a finding; its practical consequence is that comparisons of these corrections at unmatched sides measure the choice of sides.

4 Results: data-snooping tests

4.1 The studentized SPA test is oversized; the Reality Check is not

Hansen's studentized test for superior predictive ability, as Section 2 defines it, is oversized at two years of daily data and twenty strategies (Table 2). Across the nine families and both governed runs its consistent p -value rejected 4.3–12.6% of skill-less searches at a nominal 5%: 7.5% under i.i.d. normal returns, 12.5% under negative skew and 8.9% under autocorrelation in v1. It exceeded the pre-registered 6% bar in 7 of the nine families in v1. The upper p -value behaves almost identically, and StepM, whose first step uses the same studentized statistic and recentring, matched the consistent SPA test to within 0.03 percentage points in every family of both runs.

White's Reality Check, which uses the same bootstrap but does not studentize, rejected 4.5–6.4% across the same cells, and 4.5–5.7% outside the AR(1) family. Among the v1 validators it is the only one that is calibrated, or close to it, under negative skew, positive skew, fat tails, volatility clustering and correlated strategies at once.

This was not the comparison v1 was designed to make. Its pre-registration listed the two SPA p -values as candidates for the author's software and only measured the Reality Check; the pre-registered rule therefore chose the consistent SPA test, with failures in most families. Rather than switch to the Reality Check on that evidence, a second pre-registration (v1b) added it as the first candidate and repeated the run on fresh seeds. v1b confirmed the pattern (Table 2); its pre-registered rule chose the Reality Check, which missed the bar only in AR(1), $\phi = 0.2$.

The pattern does not depend on the implementation. In v2, an independent Python implementation written from the published definitions reproduced v1b's sizes of the consistent SPA test and the Reality Check in all nine families to within three Monte Carlo standard errors (prediction P1 held; the largest difference was 0.62 points).

4.2 Why: a variance estimate the bootstrap holds fixed

The studentized statistic divides each mean by $\hat{\omega}_k$, an estimate of its long-run standard deviation made once from the sample, and the resampled statistics divide by the same $\hat{\omega}_k$. The bootstrap therefore reproduces the sampling variation of the means but not that of the estimate in the denominator. Hall and Wilson [15] give the general rule this breaks: a resampled statistic should be built the same way as the observed one. The Politis–Romano estimator makes the omission costly, because at the bootstrap's block length it adds weighted sample autocovariances to the variance. Under i.i.d. normal returns its standard deviation is, to first order, 8.6% of ω at block length 8 and 14.3% at 22, against 3.1% for the sample standard deviation. A studentized mean then behaves roughly like Student t with about 68 (block 8) or 24 (block 22) degrees of freedom, while the bootstrap's critical value for the best of 20 strategies stays near the normal value of 2.80. This back-of-the-envelope account predicts sizes of 6.5% and 9.4%; it gets the ordering right and

accounts for 59% and 56% of the excess measured in experiment A, the rest coming from what it ignores (the skewness of the variance estimate’s own distribution, and the bootstrap’s noise).

Experiment A tests the account directly (Table 3). Studentizing by the sample standard deviation instead, with the bootstrap unchanged, removes most of the excess where returns are symmetric and serially uncorrelated: 5.5% under i.i.d. normal returns, 5.1% under t_4 , 5.1% under GARCH. It does not remove the excess under negative skew (10.2%), where a second effect appears: a high sample mean tends to come with a low sample standard deviation, so the studentized statistic has a heavier upper tail than any fixed denominator can reproduce. A bootstrap- t that re-studentizes every resample by its own standard deviation reproduces both effects and is calibrated under negative skew (5.4%); outside the AR(1) family its size was 5.0–6.2% (predictions P3 and P4 held and held). With the long block arch uses by default ($\lfloor \sqrt{T} \rfloor = 22$), the fixed-variance studentized test rejected 12.9% of i.i.d. normal searches and 17.8% under negative skew (P6 held).

One pre-registered prediction about the mechanism failed. P2 expected the sample-standard-deviation version to stay at or below 6% in 7 families; it did in 6, but rejected 6.4% in the block-cluster family. In that family every joint test was slightly liberal in v2, including the unstudentized ones (Reality Check 6.1%, bootstrap- t 6.2%), as were all the joint tests with only five i.i.d. normal strategies in experiment B (Reality Check 6.0%). A search of twenty strategies in four tight clusters has few effectively independent strategies, and the bootstrap’s own noise, which matters less when a maximum over many strategies dominates the statistic, is then visible.

4.3 How the excess depends on the sample and the search

The excess is a finite-sample effect: it shrinks as the sample grows and grows with the number of strategies (Table 5; prediction P7 held). Under i.i.d. normal returns the fixed-variance test rejected 9.5% of skill-less searches with one year of daily data, 7.1% with two and 6.2% with ten; with two years of data it rejected 6.9% of searches of five strategies and 9.5% of searches of a hundred. Under negative skew the same figures were 16.2%, 12.1%, 7.7%, 8.7% and 18.4%. The bootstrap- t and the Reality Check stayed near 5% across the grid (4.3–6.2% and 3.9–6.0% over all twelve cells). Hsu et al. [21] report family-wise error rates of the studentized stepwise SPA test controlled at the nominal level in simulations with 1,000 periods and between 90 and 9,000 strategies; their designs differ from ours in sample length, block length and the mix of neutral and poor strategies, and we have not reproduced them. Distinct from this finite-variance effect, Frederiksen et al. [13] show that when loss differentials have infinite variance, conventional predictive-ability tests can be severely oversized even asymptotically.

4.4 What this means for software

The open-source Python package arch [33] does not studentize its SPA statistic. In version 8.0.0 the `studentize` option changes neither the statistic nor the resampled values; the variance estimate enters only the consistent recentring threshold. On 12,000 search-and-block-length pairs in experiment D, arch’s p -values were identical with `studentize=True` and `studentize=False` on *every* pair (0 differed), and they agreed with this paper’s unstudentized consistent test to within Monte Carlo error (mean absolute difference 0.014–0.015; Table 11). Arch’s “SPA” test is thus the Reality Check with Hansen’s consistent recentring, and in the Null Zoo it stays close to its level outside the AR(1) family (4.6–6.2% for our implementation of the same computation). A pull request opened on 27 September 2026 proposes to make `studentize=True` divide by the fixed standard error, at arch’s default block length [1]. Apart from the floor at zero, which rarely matters under these null hypotheses, that is the computation measured above at 12.9% and 17.8%; we implemented it

independently from the formula and did not run the pull request’s code. Studentizing inside the bootstrap would avoid the excess.

A second, smaller issue is a matter of convention. With the statistic floored at zero, $T^{\text{SPA}} = 0$ whenever no strategy beats the benchmark in sample. If every strategy also falls below the consistent threshold, the recentred resampled statistics are mostly zero, and a p -value computed with a strict inequality, $\Pr^*(T^* > T)$, counts only the few resamples above zero: the test rejects although no strategy did better than the benchmark. This caused the second failed prediction. P9 expected the bootstrap- t version to keep its size in a design whose other strategies were clearly poor; it rejected 7.3%, as did every floored variant, while arch’s unfloored computation rejected 4.8%. An exploratory check that was not pre-registered (fresh seed 20261010, 10,000 searches) confirmed the cause: the observed statistic was zero in 49.1% of searches and every strategy fell below the threshold in 2.7%, which contributed 2.5 points of the bootstrap- t test’s 7.8%. Counting ties, $\Pr^*(T^* \geq T)$, which returns $p = 1$ whenever $T = 0$, brought it to 5.3%. The implementation behind v1, the author’s, also uses the strict inequality; in v1’s null cells, where every strategy has mean zero, a search in which every strategy falls below the threshold is too rare to matter.

4.5 Where studentizing pays off

Studentizing exists for a reason, and experiment C shows it (Table 6). When the strategies’ volatilities range from 0.5 to 2 and the skilled strategy is the least volatile, the largest mean is the skilled strategy’s in only 16.7% of searches, while the largest t -statistic is in 80.4%. The Reality Check, which ranks by mean, then rejects in 5.0% of searches with a Sharpe ratio of 2: no more than under the null. Arch’s unstudentized computation does no better (5.2%). The studentized tests find the strategy about half the time (bootstrap- t 53.1%, size-adjusted 52.4%), and the fixed-variance version is again oversized (7.6%). When the other strategies are clearly poor, the Reality Check, which recentres every strategy at its mean, is very conservative (size 0.2%, power 51.4%), and Hansen’s consistent recentring raises power to 87.3% at a size of 4.8% (unstudentized) (P8 held). The lesson is not to give up studentizing or recentring, which is why Hansen proposed them, but to studentize inside the bootstrap and to count ties.

4.6 Power

Where the strategies’ volatilities are equal, as in the nine families, studentizing buys no power: at a Sharpe ratio of 2 the bootstrap- t version and the Reality Check had similar size-adjusted power under i.i.d. normal returns (53.4% and 53.2%; Table 10). Skew pulls them apart in opposite directions: under negative skew the Reality Check was ahead (54.9% against 46.7%), under positive skew the bootstrap- t (57.8% against 49.1%). The fixed-variance test’s apparent power advantage (raw 56.9% in v1, against 53.5% for the Reality Check) disappears once its excess size is removed (50.2% against 53.2%).

4.7 The pre-registered predictions

Of the 9 predictions registered before the v2 run, 7 held. Table 13 lists all of them with their outcomes. The two that failed, P2 and P9, are reported above with their explanations; the explanation of P9 comes from an exploratory check made after the run and should be read as such.

5 Discussion

Implications for practice. Six suggestions follow from the tables, within the limits below.

1. *Studentize inside the bootstrap, or not at all.* A studentized data-snooping test whose resampled statistics reuse the sample’s variance estimate is oversized at two years of daily data, more so with long blocks, many strategies or negative skew. Re-studentizing every resample keeps the test calibrated and keeps the power studentizing buys when volatilities differ. When strategies have similar volatilities, the Reality Check is a calibrated and simple alternative.
2. *Recentre consistently and count ties.* Hansen’s consistent recentring makes the test far more powerful when many variants are clearly poor; computing the p -value as $\Pr^*(T^* \geq T)$ stops a floored statistic from rejecting when no strategy beats the benchmark.
3. *Check the selected strategy’s returns for autocorrelation* before any multiple-testing adjustment. Lo’s factor removes the excess in the autoregressive case it assumes; kernel and block-bootstrap corrections reduce it without that assumption, at a cost in size at this sample length.
4. *Compare corrections at matched sides and with the number of resamples stated.* Otherwise the comparison measures conventions: the haircut’s conservatism comes from its two-sided convention, and a bootstrap’s power from its number of resamples.
5. *Treat $\text{DSR} \geq 0.95$ as the demanding rule it is.* It asks whether the best strategy beats the expected best of N skill-less ones, not whether it has any skill, and over two years of daily data it passes few strategies with a true Sharpe ratio of 2.
6. *When returns are markedly skewed or fat-tailed, do not trust a single-series correction:* none in the benchmark is calibrated across negative skew, positive skew and fat tails, in part because selection itself distorts the moments of the chosen series. The Reality Check and the bootstrap- t SPA test, which resample all strategies together, were.

Limitations. The nine families change one feature at a time; real returns combine several, and the only serial dependence studied is AR(1), the Lo factor’s best case. The main grid covers one search shape ($N = 20$, $T = 504$) and one strongly skilled strategy; experiment B varies T and N under two families only, and real searches often contain many weakly skilled strategies. The block length of every resampling test is fixed by rule rather than chosen from the data [30]. The bootstrap- t studentizes each resample by its own standard deviation, which ignores serial dependence; a block-based studentization [14] would be the natural choice for autocorrelated returns, and is untested here. The deflated Sharpe ratio is scored only at its 0.95 point, not as a probability. Experiment C uses i.i.d. normal returns, so it isolates the role of studentizing and recentring but not their interaction with skew. Finally, a benchmark designed by the author of one of the scored methods, whose runs also chose a default for the author’s own software, cannot be neutral in the sense of Bartoš et al. [7]. We pre-register the runs and publish every generator, validator, seed and result file, but independent replication with independently chosen families would be more convincing than anything we can do ourselves.

Next versions. The benchmark is versioned so that it can grow without rewriting its history. Natural additions are families that combine features (skewed and autocorrelated returns) or have non-AR dependence, alternatives with many weakly skilled strategies, data-driven block lengths,

block-studentized bootstrap tests, and an estimate of the effective number of trials for correlated searches.

6 Conclusion

The corrections practitioners use for backtest overfitting and data snooping keep their promises in the setting they were derived for and fail in predictable ways outside it. The most consequential failure found here is the most technical: Hansen’s studentized SPA test, recommended over the Reality Check for its power, is oversized at the two-year horizon of strategy research because its bootstrap reuses a noisy variance estimate, and the arch implementation stays close to its level only because it does not do what its option says. Both are easy to repair. Autocorrelation remains the most damaging violation for the single-series corrections, and several apparent differences between methods come from conventions—sides, thresholds, numbers of resamples—rather than from the methods. An open, pre-registered benchmark with known ground truth makes such statements checkable, and anyone can rerun or extend it from the published code and seeds.

Code and data availability

All code, result files, pre-registrations and the source of this paper are at <https://github.com/arhancanli/null-zoo> (MIT licence); version 1 of the paper is archived at <https://doi.org/10.5281/zenodo.23018988>. The JavaScript benchmark in `src/` is copied unchanged from <https://github.com/arhancanli/canlicapital> at commits 4f7838f8db35 (v0) and ea3985981ec8 (v1, v1b), where it is maintained; the published v0 result file (SHA-256 e8287f19cf25a7e0858ab12d19e2c2154f1068c5c32459123d68ec3a7cd3ce0f) reproduces byte for byte. The independent Python implementation, its instrument tests, the v2 pre-registration (commit 7658b3b46ddb), the run and scoring scripts and their outputs are in `analysis/v2/`. Every number in the text and tables is generated from the result files by `analysis/tables.mjs`, and the paper reads each one by key, so none is typed by hand. Continuous integration checks the instrument tests, the byte-for-byte reproduction of v0 and that the committed tables match the result files.

Competing interests

The author designed the luck-equivalent-trials statistic scored here and maintains open-source implementations of several of the scored methods, including the data-snooping tests; the v1 and v1b runs were also used to choose a default test for that software. No external funding was received.

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Table 7: Size of the remaining v1 validators, and of every joint test in the confirmation run v1b (percent, 10,000 searches per cell).

Family	HC 2s	HC-Lo	Holm	Oracle	SPA _u v1b	StepM v1b
IID normal	<i>2.2</i>	5.0	4.8	5.6	7.5	7.7
Student t_4	<i>2.3</i>	5.1	4.6	5.4	7.5	7.6
Skew -1.32	<i>4.4</i>	8.5	8.1	5.9	12.3	12.6
Skew $+1.32$	<i>1.3</i>	<i>3.3</i>	<i>2.9</i>	4.5	<i>4.0</i>	<i>4.3</i>
GARCH(1,1)	<i>2.3</i>	4.9	4.9	5.1	6.2	6.3
AR(1), $\phi = 0.2$	11.9	5.3	19.3	5.7	8.9	9.1
Two vol. regimes	<i>2.7</i>	5.1	5.0	5.3	6.0	6.1
Correlated, $\rho = 0.5$	<i>1.9</i>	<i>3.7</i>	<i>3.5</i>	5.4	6.9	6.9
Clusters, $\rho = 0.8$	<i>1.6</i>	<i>3.4</i>	<i>3.3</i>	5.1	6.9	7.0

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it. HC 2s: the haircut with the two-sided p -values its authors define, so its nominal one-sided level is 2.5%.

Table 8: Raw power at a true annualized Sharpe ratio of 1 (percent), Null Zoo v1.

Family	Oracle	LET	LET-Lo	DSR	Boot	HAC- t	RC	SPA _c
IID normal	41.2	12.4	13.0	0.4	12.1	13.7	13.1	16.8
Student t_4	41.9	13.2	13.8	0.6	14.8	14.3	12.4	17.0
Skew -1.32	42.5	16.8	17.3	0.5	11.7	18.4	14.0	23.6
Skew $+1.32$	41.1	9.7	10.2	0.4	13.3	10.8	12.5	11.6
GARCH(1,1)	42.7	13.6	13.7	0.5	12.7	14.3	12.4	16.7
AR(1), $\phi = 0.2$	33.4	29.9	10.1	0.9	28.4	13.4	11.6	15.6
Two vol. regimes	42.7	13.8	14.2	0.6	13.6	14.2	11.4	15.8
Correlated, $\rho = 0.5$	41.5	10.1	10.4	4.0	9.6	11.1	14.6	16.4
Clusters, $\rho = 0.8$	41.3	11.0	11.4	0.7	10.7	12.1	16.0	18.1

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A Additional tables

Table 7 gives the size of the validators not shown in the main tables and of the joint tests in the confirmation run v1b. Tables 8 and 9 give raw power at Sharpe ratios of 1 and 3. Table 10 gives the power of the SPA variants of experiment A, Table 11 the check of arch itself (experiment D), Table 12 the paired bootstrap-resolution run, and Table 13 the pre-registered predictions of v2 and their outcomes.

Table 9: Raw power at a true annualized Sharpe ratio of 3 (percent), Null Zoo v1.

Family	Oracle	LET	LET-Lo	DSR	Boot	HAC- t	RC	SPA _c
IID normal	99.4	93.0	92.8	50.6	92.2	93.0	93.1	93.6
Student t_4	99.5	92.1	92.0	49.9	90.6	92.2	91.9	92.9
Skew -1.32	99.3	91.3	91.2	41.1	86.8	91.4	93.6	93.1
Skew $+1.32$	99.8	94.3	94.1	60.1	95.4	94.3	92.0	94.0
GARCH(1,1)	99.4	92.4	92.3	52.3	91.5	92.5	92.3	93.1
AR(1), $\phi = 0.2$	97.2	90.1	75.9	37.7	89.3	79.4	78.4	80.6
Two vol. regimes	99.3	91.6	91.5	53.1	90.6	91.4	91.2	91.6
Correlated, $\rho = 0.5$	99.5	92.4	92.1	65.6	91.6	92.5	94.7	95.1
Clusters, $\rho = 0.8$	99.5	93.1	92.9	59.9	92.1	93.1	95.5	95.5

Table 10: Power (raw / size-adjusted) of the SPA variants at a true annualized Sharpe ratio of 2, Null Zoo v2 experiment A.

Family	SPA _c	SPA _c -SD	SPA _c -BT	RC	SPA _c -U
IID normal	57.2 / 50.9	54.6 / 53.0	53.9 / 53.4	53.8 / 53.2	54.2 / 53.2
Student t_4	57.5 / 52.6	54.8 / 54.6	54.8 / 54.5	51.6 / 52.7	52.0 / 53.1
Skew -1.32	60.9 / 44.1	58.6 / 46.8	48.1 / 46.7	55.7 / 54.9	56.1 / 54.8
Skew $+1.32$	53.0 / 55.7	49.6 / 58.2	59.2 / 57.8	49.7 / 49.1	50.0 / 49.4
GARCH(1,1)	58.4 / 54.6	55.3 / 54.9	55.0 / 54.3	50.7 / 50.7	51.1 / 50.8
AR(1), $\phi = 0.2$	42.8 / 31.6	39.8 / 34.2	39.0 / 34.2	38.7 / 34.8	39.2 / 34.8
Two vol. regimes	58.1 / 56.1	54.7 / 54.9	54.8 / 54.8	49.3 / 50.4	49.6 / 50.5
Correlated, $\rho = 0.5$	59.9 / 54.8	58.8 / 56.5	58.1 / 56.5	58.2 / 56.5	58.4 / 56.7
Clusters, $\rho = 0.8$	61.9 / 53.8	60.8 / 55.6	60.2 / 55.6	60.1 / 56.3	60.3 / 56.1

Table 11: arch 8.0.0 on Null Zoo searches (experiment D, 2,000 null searches per family): size of arch’s consistent and upper p -values and of this paper’s implementation on the same searches, at block length 8 and at arch’s default (22).

Family	arch _c (8)	SPA _c -U (8)	arch _u (8)	RC (8)	arch _c (22)	SPA _c -U (22)	SPA _c (22)
AR(1), $\phi = 0.2$	7.8	7.6	7.7	7.4	7.4	7.7	13.9
IID normal	5.7	6.0	5.6	5.8	6.5	6.5	12.7
Skew -1.32	5.1	5.0	5.0	4.7	5.5	5.5	17.9

Bold: above the 99% Monte Carlo band around 5% (3.74–6.26 for 2,000 searches); italic: below it. Searches on which arch’s p -values differed between `studentize=True` and `studentize=False`: 0 of 12,000 search-and-block pairs.

Table 12: Bootstrap resolution: the same 10,000 searches per cell scored by the single-series bootstrap with 400 and with 1,999 resamples (v0 design, robustness generator). Size and power in percent.

Family	LET size	Boot ₄₀₀ size	Boot ₁₉₉₉ size	LET power	Boot ₄₀₀ power	Boot ₁₉₉₉ power
IID normal	4.59	4.28	4.50	52.6	46.7	50.9
Student t_4	4.80	5.86	6.24	54.1	49.0	53.9
Skew -1.32	8.02	4.89	5.26	54.9	40.9	46.0

Table 13: The nine predictions registered before the v2 run (analysis/v2/prereg-v2.json, commit 7658b3b46ddb) and whether each held, as scored by analysis/v2/evaluate_v2.py.

ID	Prediction	Outcome
P1	Replication: in every one of the nine families, the null size of spa_c and of rc in experiment A differs from v1b's spa_consistent and reality_check by at most three combined Monte Carlo standard errors.	held
P2	Variance noise: spa_c_sd has size at most 0.06 in iid_normal, student_t4, skew_positive, garch, regimes, correlated_trials and block_cluster (experiment A).	did not hold
P3	Skew: in skew_negative, spa_c_sd has size above 0.08 and spa_c_boot_t has size at most 0.065 (experiment A).	held
P4	Bootstrap-t: spa_c_boot_t has size at most 0.065 in all eight families other than ar1 (experiment A).	held
P5	arch 8.0.0: spa_c_unstud has size at most 0.065 in all eight families other than ar1 (experiment A), and in experiment D arch's p-values are identical under studentize=True and studentize=False on every search.	held
P6	Default block: in iid_normal, spa_c_b22 has size at least one percentage point above spa_c at block 8 (experiment A).	held
P7	Sample size and strategies: in iid_normal, the size of spa_c at n = 2520 is at least one percentage point below its size at n = 252, and at k = 100 at least one percentage point above its size at k = 5 (experiment B).	held
P8	Studentizing pays off: in heterogeneous_volatility, the size-adjusted power of spa_c_boot_t exceeds that of rc by at least 0.05; in poor_alternatives, the raw power of spa_c_boot_t exceeds that of rc by at least 0.05 (experiment C).	held
P9	Recentring: in poor_alternatives under the null, rc has size below 0.03 and spa_c_boot_t has size at most 0.065 (experiment C).	did not hold